

Efficiency of a simple baseline

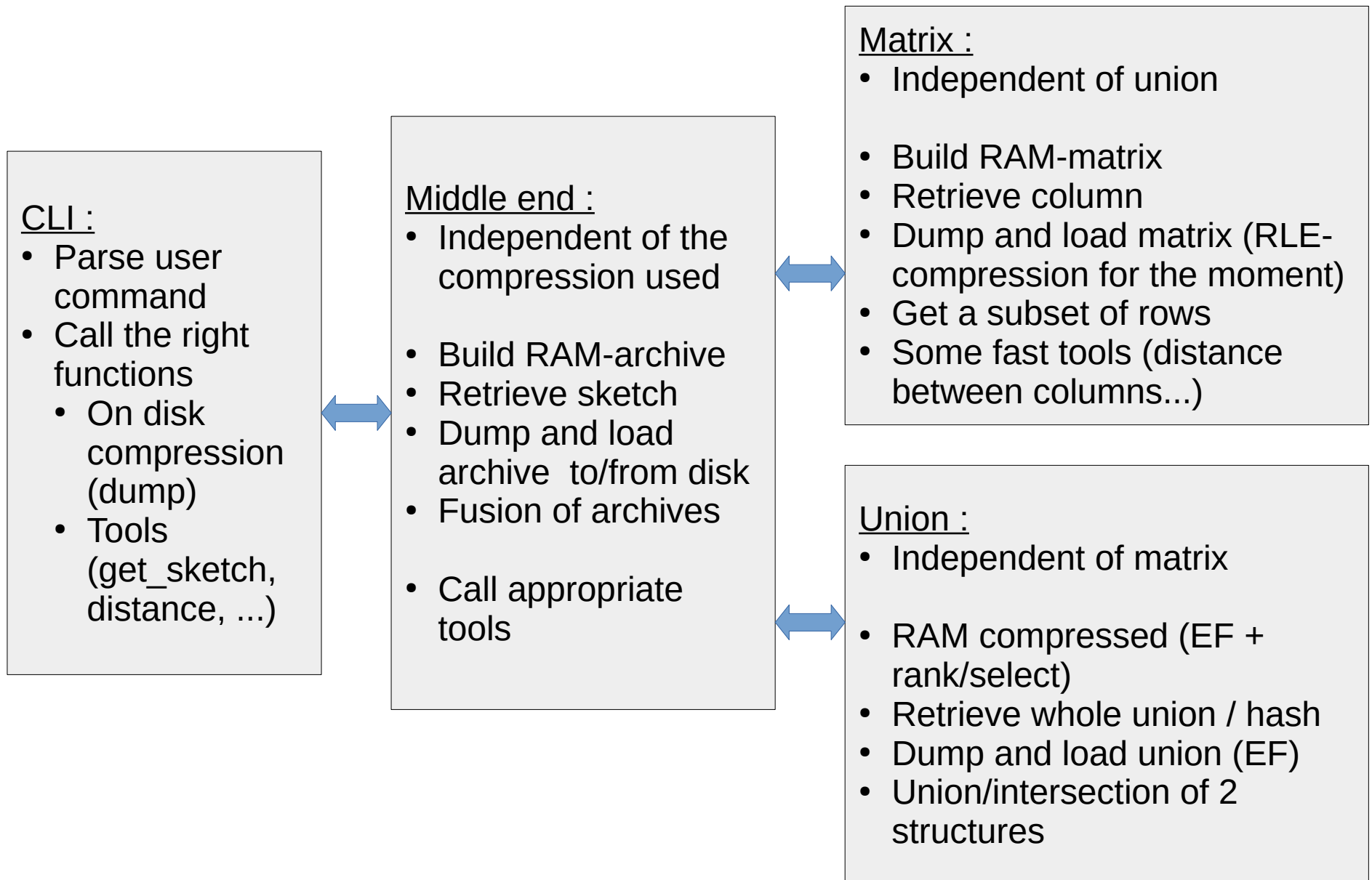
Marie Picard

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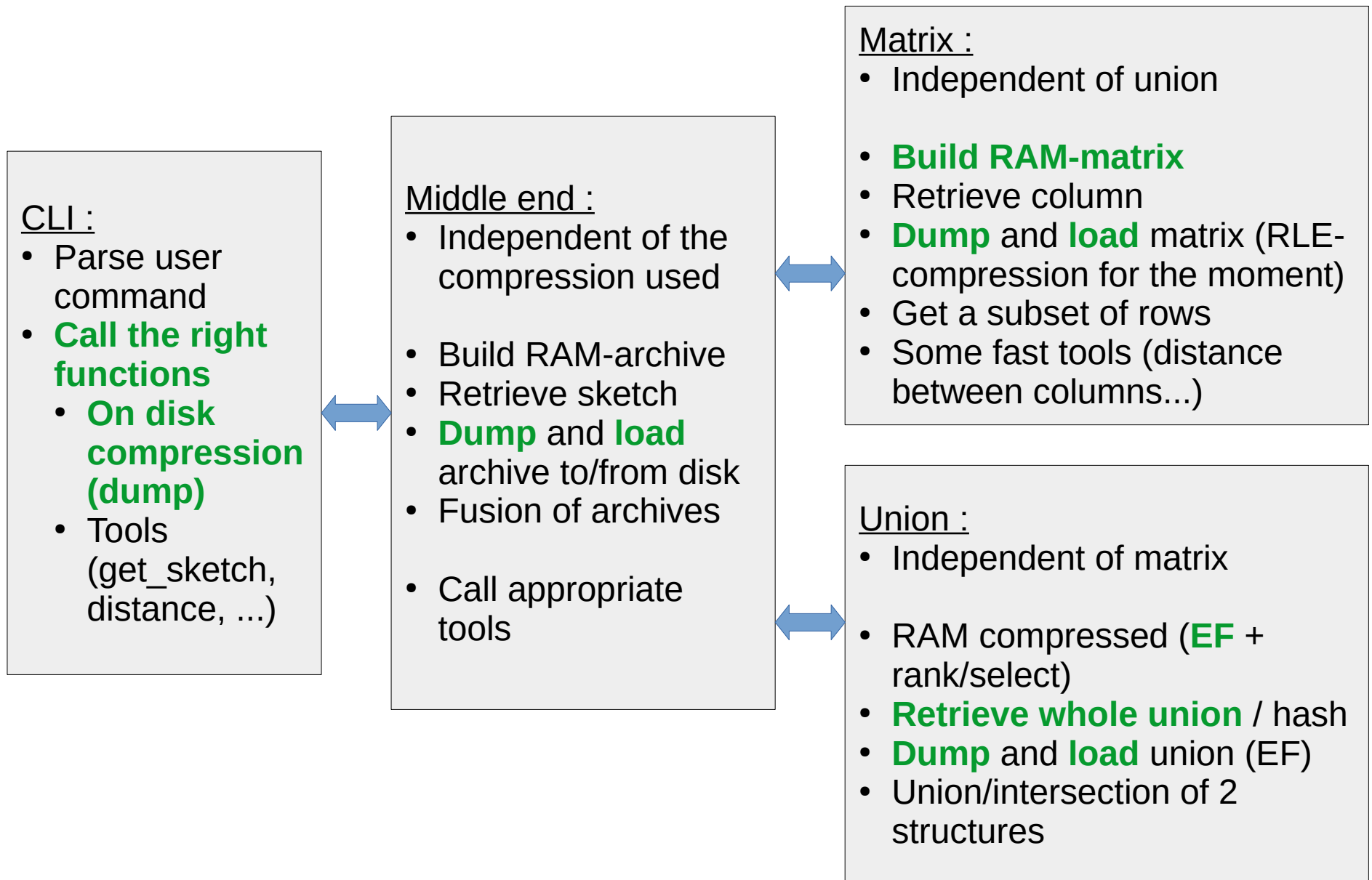
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March 13, 2026

Quick recap : code design

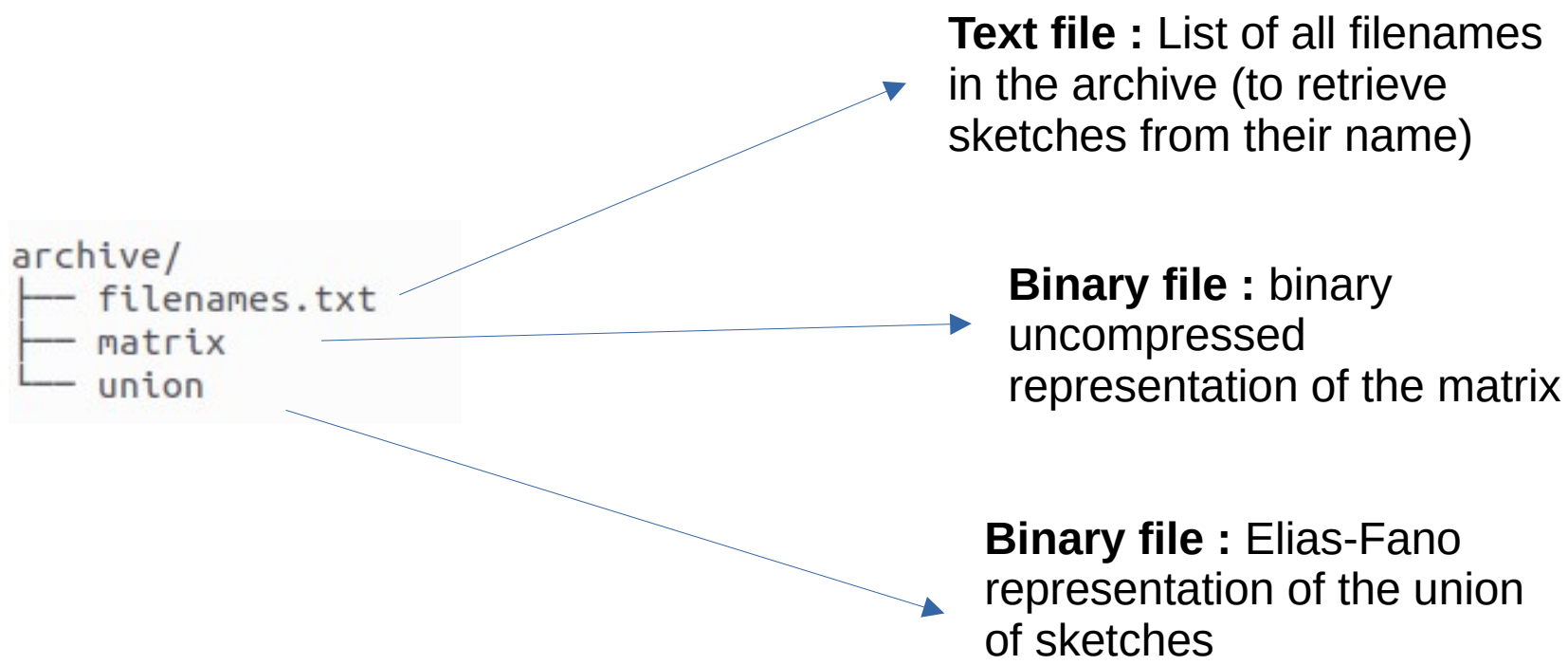


Quick recap : code design



Structure of output

```
archive/  
├── filenames.txt  
├── matrix  
└── union
```

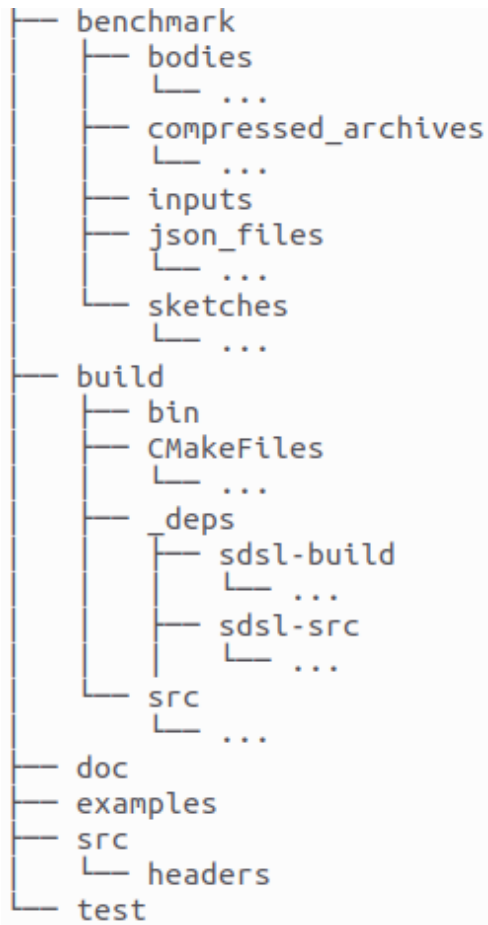


Text file : List of all filenames in the archive (to retrieve sketches from their name)

Binary file : binary uncompressed representation of the matrix

Binary file : Elias-Fano representation of the union of sketches

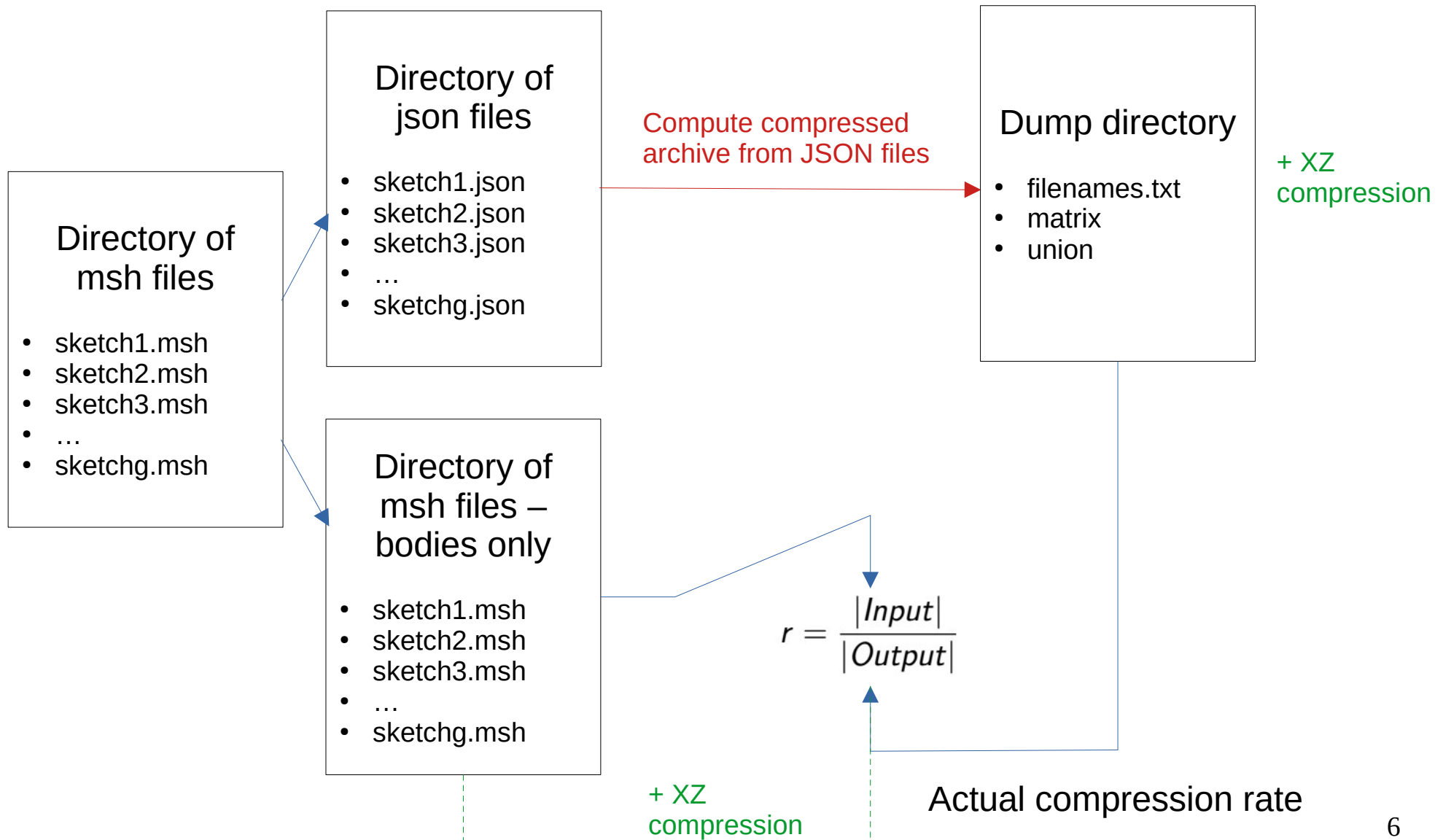
Code structure and CLI



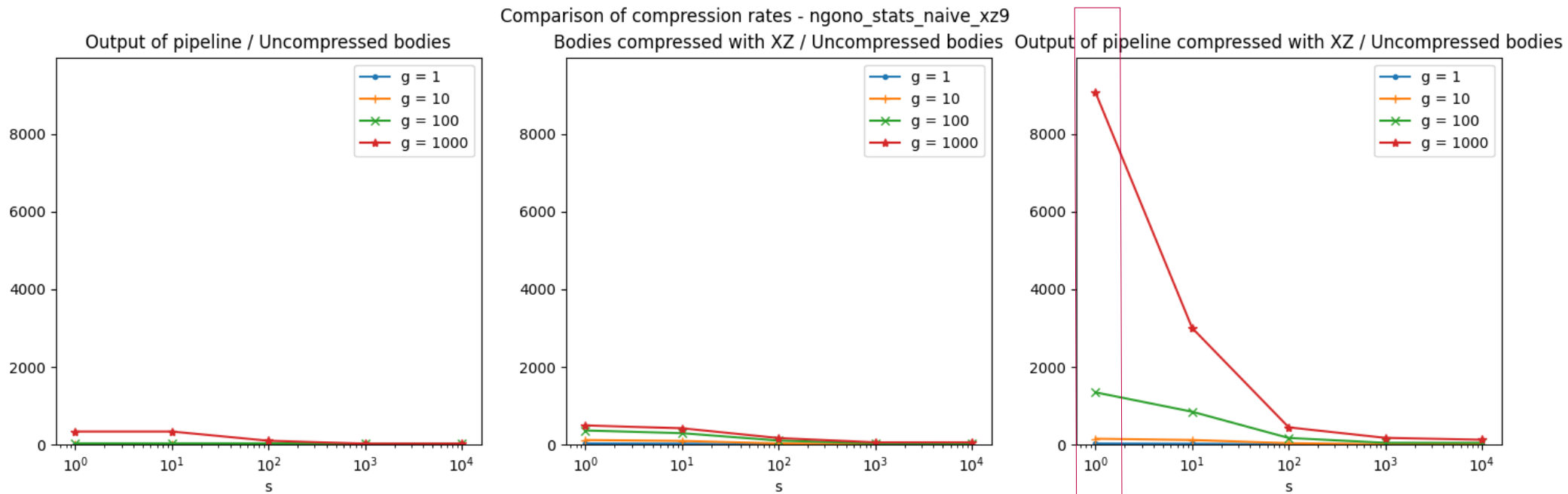
- **Input** : directory of JSON files*
- **Output** : directory of filenames + union + matrix
- Only **current operation** : store archive on disk

*not .msh for architecture-related reasons

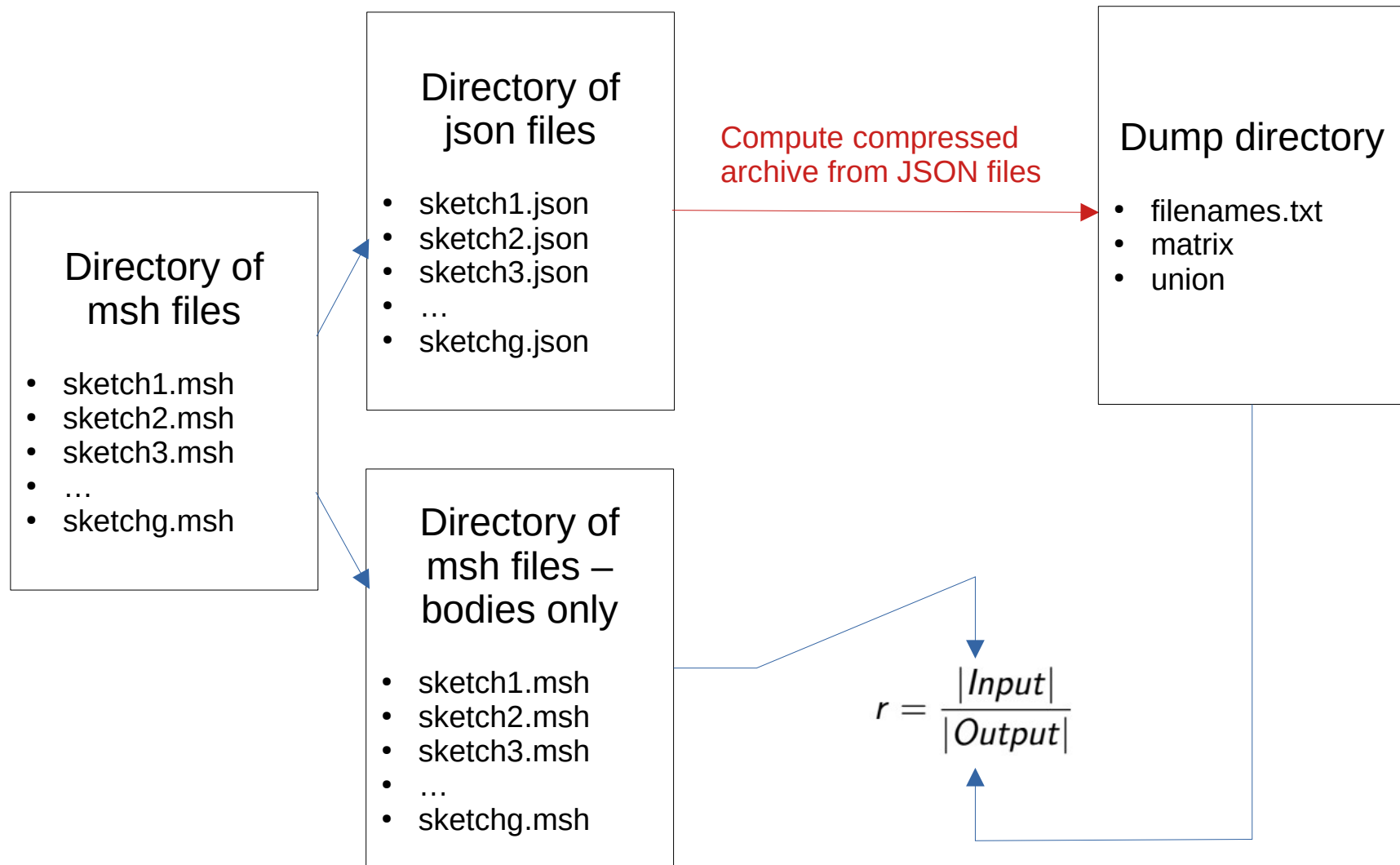
Benchmark pipeline



Actual compression rates



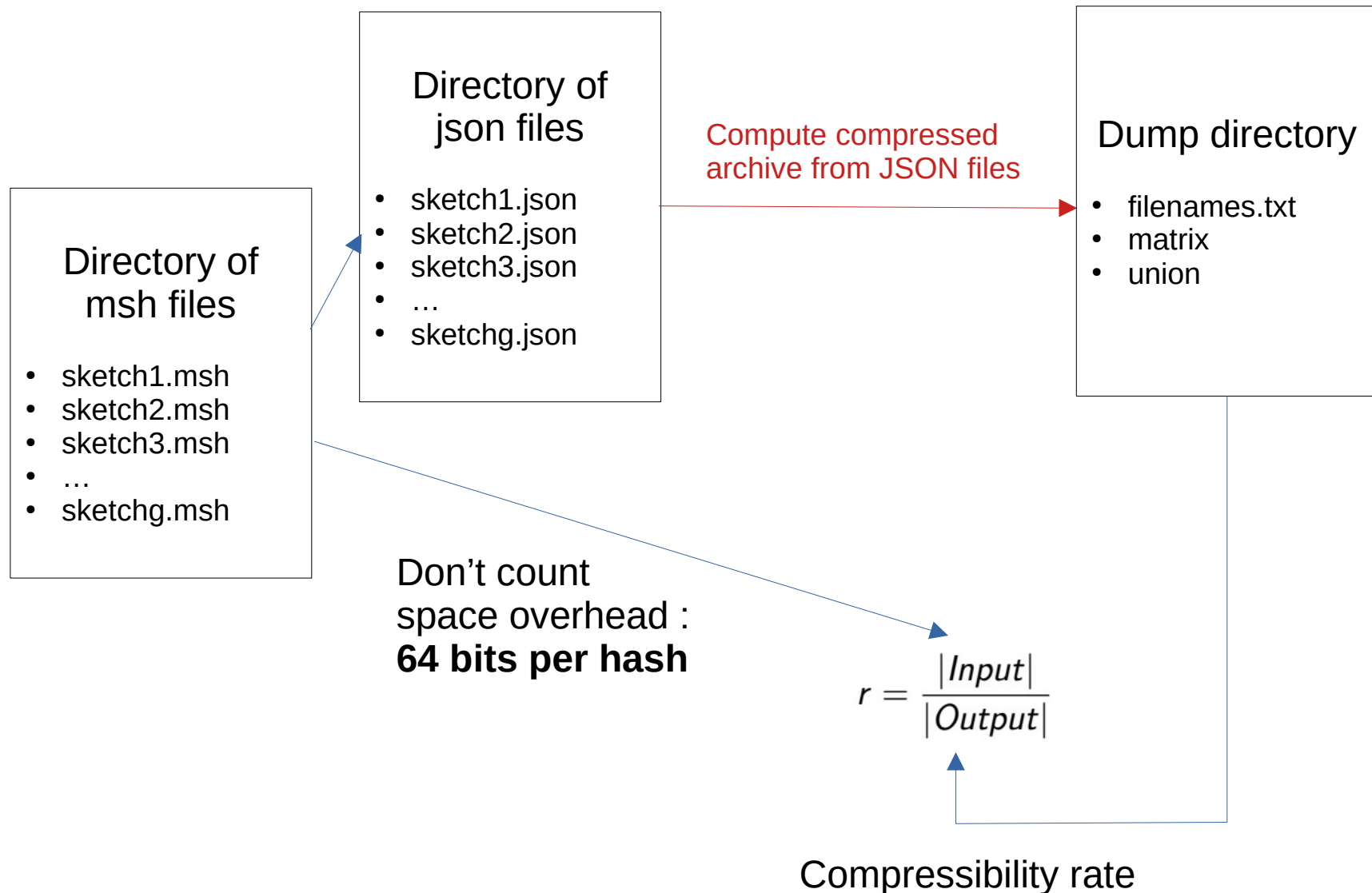
Benchmark pipeline



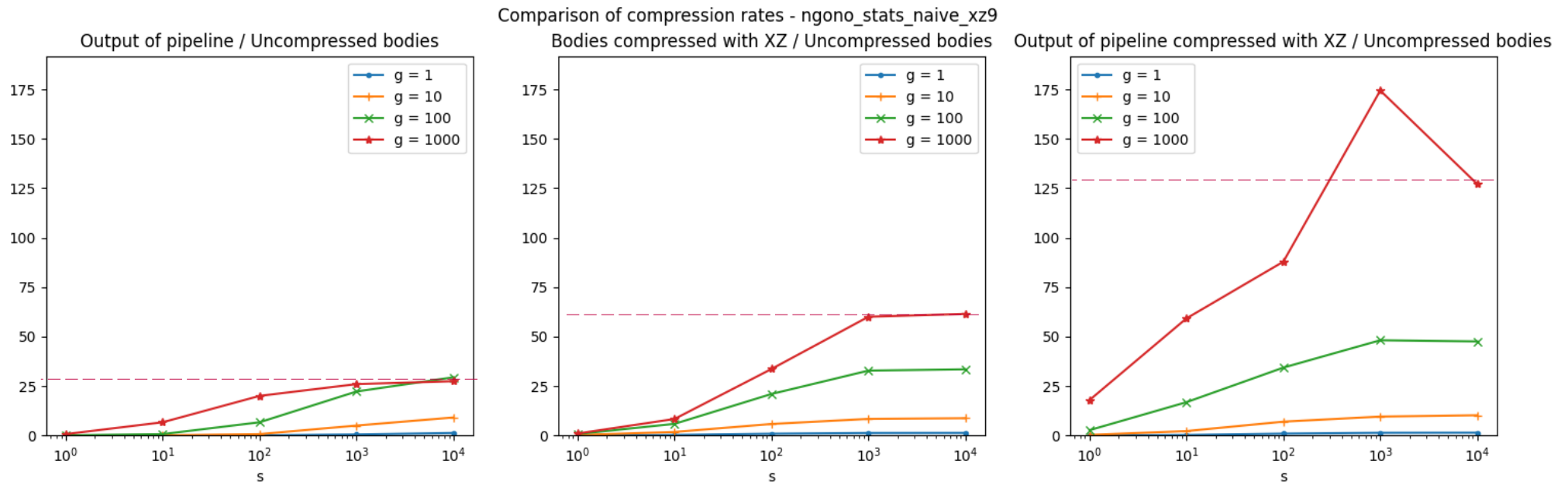
Space overhead for file &
directory definitions

Actual compression rate

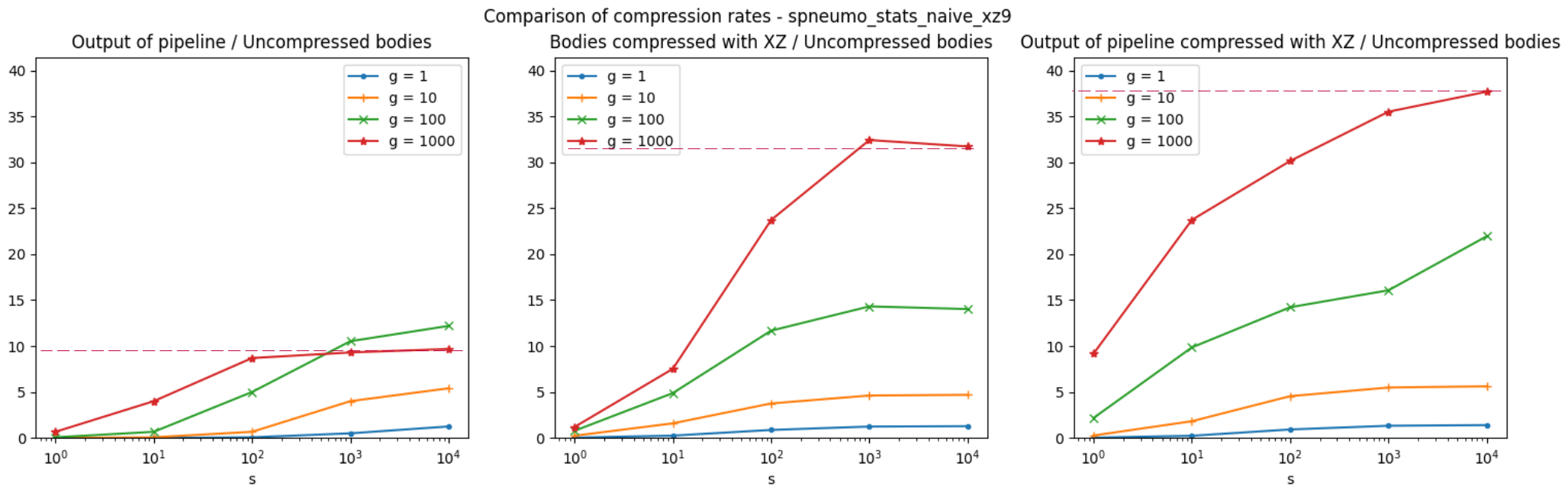
Benchmark pipeline



Compressibility rates



Compressibility rates



Conclusion

Conclusions :

- Great compression rate, even better with XZ
- Compression rate improves with $g \rightarrow$ contrary to other compression tools (like GZIP, and on a smaller scale XZ)

To do :

- Start a draft of the report
- Start working on a better RAM-access version
- Try other compression tools than XZ to improve compression of the matrix
- Find a way to make it parse .msh files instead of JSON